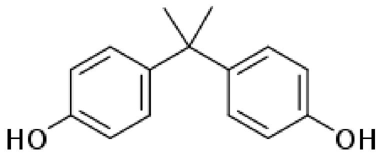
Chemical Hazards: Bisphenol-A, Mercury

Chemical hazard

A chemical hazard is any non-biological substance that has the potential to cause harm to life or health. It can include any single or combination of toxic chemical, biological, or physical agents in the environment, resulting from human activities or natural processes, that may impact the health of exposed subjects. This can include pollutants such as heavy metals, pesticides, biological contaminants, toxic waste, industrial and home chemicals. Chemical hazards and toxic substances pose a wide range of health hazards (such as irritation, sensitization, and carcinogenicity) and physical hazards (such as flammability, corrosion, and explosibility).

In this section, we will look at bisphenol A and problems associated with its poisoning in the human body. We will also look at contamination due to heavy metals, specifically mercury.

Bisphenol A



Bisphenol A (BPA) is a chemical compound primarily used in the manufacturing of various plastics. It is a colourless solid which is soluble in most common organic solvents but has very poor solubility in water. BPA's largest single application is as a co-monomer in the production of polycarbonates, which accounts for 65-70% of all BPA production. The manufacturing of epoxy resins and vinyl ester resins account for 25-30% of BPA use.

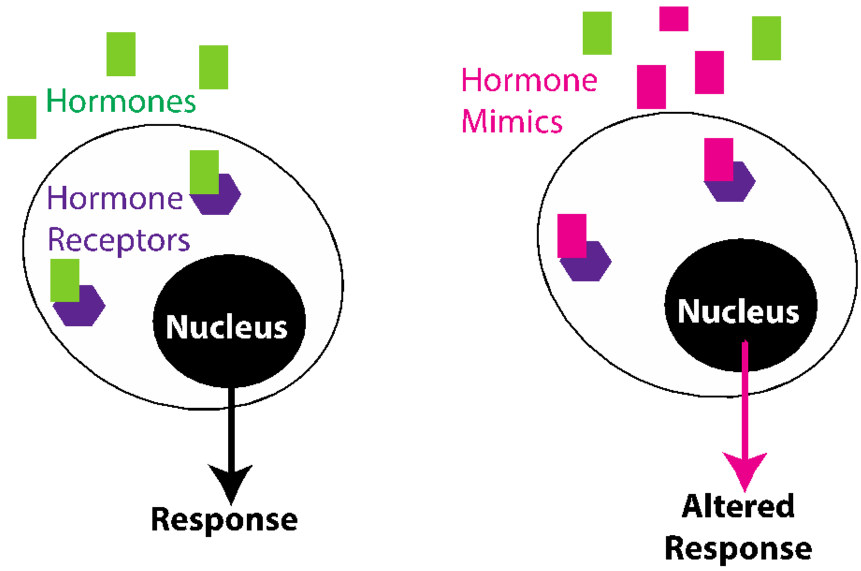
Products containing Bisphenol A

BPA is found in polycarbonate plastics and epoxy resins. Polycarbonate plastics are often used in containers that store food and beverages, such as water bottles. Epoxy resins are used to coat the inside of metal products, such as food cans, bottle tops and water supply lines. Common products that may contain BPA include:

- Items packaged in plastic containers
- Baby bottles
- Canned foods
- Toiletries
- Menstrual products
- Thermal printer receipts
- CDs and DVDs
- Household electronics
- Eyeglass lenses
- Sports equipment
- Dental filling sealants

Mechanism of action of BPA in the human body

BPA binds to both nuclear estrogen receptors (ERs), ERα and ERβ, activating them. It can mimic as well as antagonize estrogen, indicating that it is a selective estrogen receptor modulator (SERM) or partial agonist. It also acts as an antagonist of the androgen receptor (AR) at high concentrations.

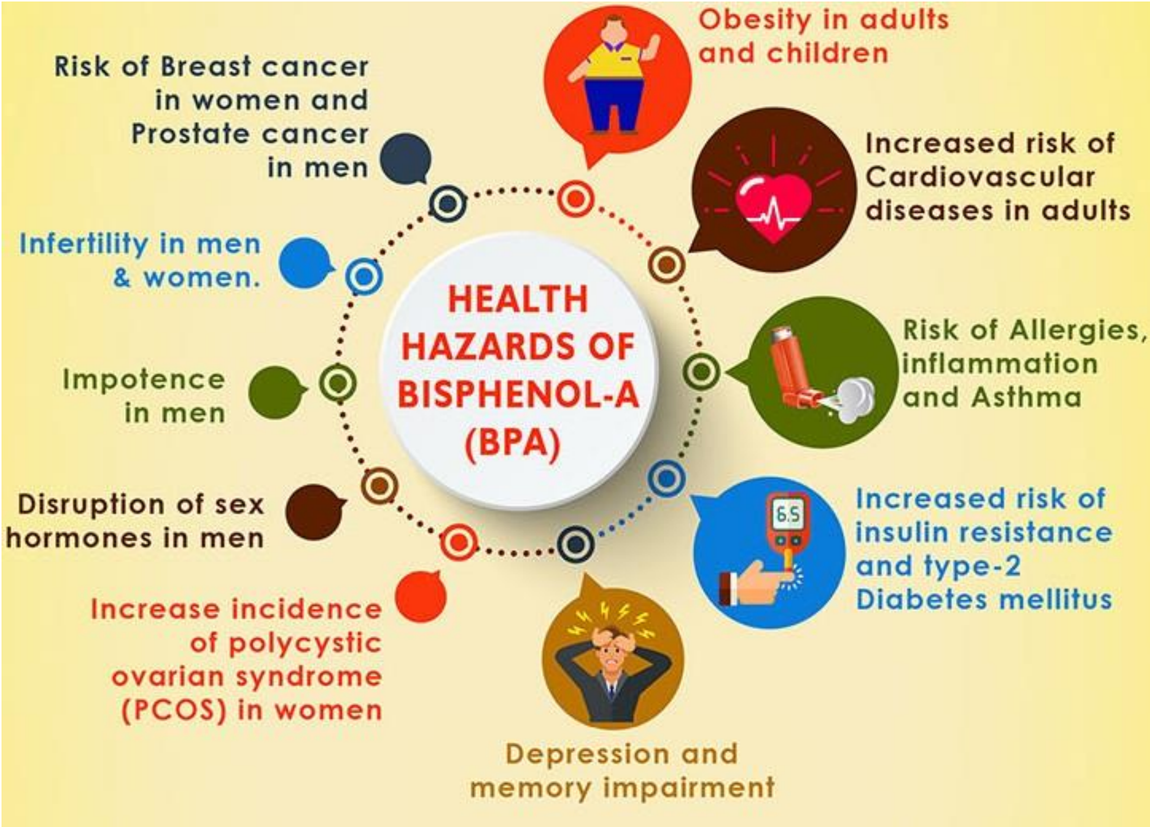


Schematic representation of the mechanism of action of BPA in the human body

Health problems associated with BPA

BPA has been linked to causing reproductive, immunity, and neurological problems, as well as an increased likelihood of Alzheimer's, childhood asthma, metabolic disease, type 2 diabetes, and cardiovascular disease.

- Several neurological health issues have been observed during pregnancy and development, like reduced lung capacity, wheezing and asthma after birth, leading to ban in the use of BPA in baby bottles.
- Studies have linked BPA and obesity; BPA exposure modifies insulin sensitivity and insulin release without affecting weight.
- Other endocrine-related disorders include infertility, polycystic ovarian syndrome (PCOS) and precocious puberty.
- BPA also disrupts thyroid function, binding to thyroid hormone receptor, and studies have linked BPA with increased TSH (Thyroid stimulating hormone).
- BPA exposure can lead to prostate cancer in men.



Summary of some health problems associated with BPA poisoning in the human body

Sources of BPA contamination

BPA can get in our body through eating or drinking foods heated in plastics; eating or drinking foods stored in metal cans (canned foods) or plastics (take-out containers); and touching cash register receipts. The major points of entry of bisphenol A into our body are summarized below.

- Major human exposure to BPA is diet, via ingestion of contaminated food and water.
- Plastics leach BPA when cleaned with harsh detergents, or when they contain acidic or high-temperature liquids.
- BPA-based resin coatings in older water pipes can leach BPA.
- Several uses of BPA in digital media, electrical and electronic equipment, sports safety equipment, electrical laminates in printed circuit boards, composites, paints and adhesives can also lead to exposure.
- Bioaccumulation in water bodies, aquatic plants and organisms can result in toxicity.
- BPA is also found in high concentrations in thermal and carbonless copy paper, used for printing receipts, airline tickets etc., and can be absorbed into body through skin.

Environmental effects of BPA

Even though BPA has a short half-life (4.5 days in soil and water, < 1 day in air), its ubiquity makes it an important pollutant. It has a low rate of evaporation from water and soil, which creates problems despite its biodegradability. It interferes with nitrogen fixation at the roots of certain leguminous plants. BPA affects growth, reproduction and development in aquatic organisms, especially fish, with endocrine-related effects observed in fish and other aquatic invertebrates, amphibians and reptiles. It also impacts reproduction in terrestrial animals and insects, impairing development and inducing genetic aberrations.

Steps to limit BPA contamination

Heavy metal poisoning

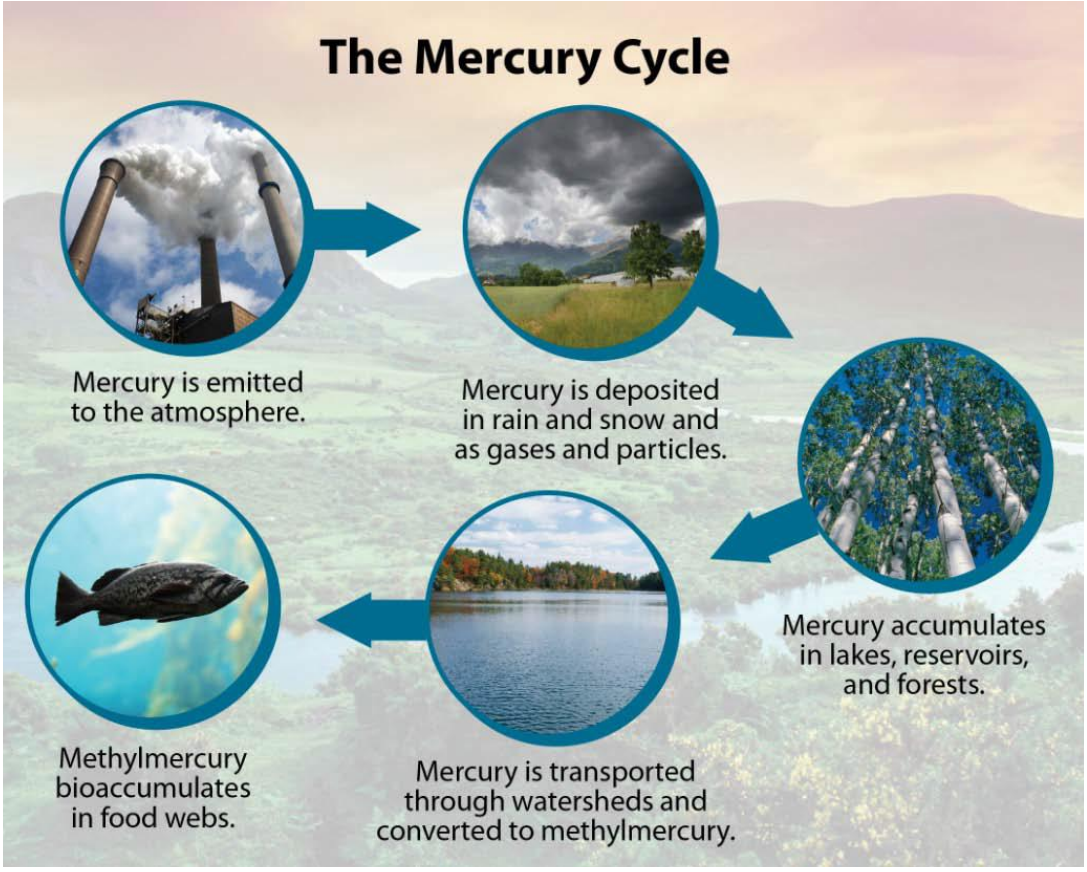
Heavy metal poisoning refers to when excessive exposure to a heavy metal affects the normal function of the body. Examples of heavy metals that can cause toxicity include lead, mercury, arsenic, cadmium, and chromium. Exposure may occur through the diet, from medications, from the environment, or in the course of work or play. Heavy metals can enter the body through the skin, or by inhalation or ingestion. Toxicity can result from sudden, severe exposure, or from chronic exposure over time.

Mercury

Mercury is a heavy metal belonging to the transition element series in the periodic table. It exists in nature in three forms: elemental, organic and inorganic, each with its own profile of toxicity. It is a liquid at room temperature; it has high vapour pressure and is released into the environment as mercury vapour. Its most commonly occurring oxidation states are +1 +2. Methylmercury is the most frequently encountered organic compound found in the environment, formed as a result of methylation of inorganic mercuric forms of mercury by microorganisms found in soil and water.

Mercury in the Environment

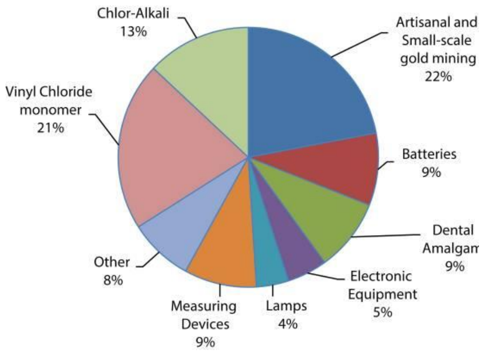
Mercury is a widespread environmental toxicant and pollutant, inducing severe alterations in the body and a wide range of adverse health effects. It is ubiquitous in the environment, therefore, making it difficult for plants, animals and humans alike to avoid exposure.



Schematic representation of the different forms of mercury in the environment. *[Source: https://webcam.srs.fs.fed.us/impacts/mercury/index.shtml]*

Uses of Mercury

- Electrical industry (switches, thermostats, batteries)
- Dental fillings
- Industrial processes (production of caustic soda)
- Nuclear reactors
- Anti-fungal agents for wood processing
- Solvent for reactive and precious metals



Sources of Mercury Poisoning

- Major sources of exposure to mercury are through accidents, environmental pollution, food contamination, dental care, preventive medical practices, industrial and agricultural operations such as non-ferrous metal production, cement production etc.
- Major sources of chronic low level mercury exposure are through dental amalgams (50% elemental mercury) and fish consumption.
- Mercury exposure can also occur by inhaling contaminated air, and improper use/disposal of mercury-containing objects after spills or disposal of fluorescent lamps.
- Human activities that can result in Mercury release into environment: burning of coal (half of atm. mercury) and gold mining.

- Mercury enters water either through Earth's crust or through industrial pollution, which are methylated by algae and bacteria in the water, which then bioaccumulates in fish, and eventually into humans.
- Two most highly absorbed species: Hg(0) and methyl mercury.

Mechanism of Mercury in the Human Body

- Elemental mercury vapour is highly lipophilic and effectively absorbed through lungs and tissues lining the mouth.
- After mercury enters the blood, it rapidly passes through cell membranes, including both the blood-brain-barrier (BBB) and placental barrier (PB).
- Within the cell, it is oxidized to its highly reactive +2 state.
- Methyl mercury, from fish, is readily absorbed in the gastrointestinal tract because of its lipid solubility, and can cross the BBB and PB.
- Once absorbed, mercury has a very low excretion rate, accumulating in kidneys, neurological tissues and liver, resulting in gastrointestinal toxicity, neuro and nephrotoxicity.

Adverse Health Effects of Mercury

- Brain is the target organ for mercury, but it can also impair any organ leading to malfunctioning of nerves, kidney and muscles.
- Mechanism: mercury binds to free thiol groups (cysteine residues).
- Symptoms depend on the type, dose and duration of exposure, including peripheral neuropathy, skin discolouration (pink), swelling and desquamation (shedding or peeling of skin).
- Mercury is neurotoxic, responsible for microtubule destruction, mitochondrial damage, lipid peroxidation etc.
- High-levels of exposure to mercury: Minamata disease.
- Symptoms: acrodynia (pink disease) skin becomes pink and peels, kidney problems, decreased intelligence.

Treatment and Prevention

- Mercury poisoning can be reduced by eliminating/reducing exposure to mercury and related compounds.
- Powdered sulfur may be applied in case of a spill, resulting in a solid compound that can be easily disposed off of.

Treatment

- First step is decontamination, disposal of clothes, washing skin with soap and water, flushing eyes with saline as needed.
- Chelation therapy with DMSA and other sulfur-based compounds are effective for inorganic mercury poisoning.
- DMSA can be used against severe mercury poisoning.